Structure-emergent vs. number-convention: decomposing an autonomous chat agent's emit policy

A 4-SUPPORTED / 4-FALSIFIED separation across eight pre-registered falsifiers (UNIVERSE/H_632–H_639)

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Abstract

Is the emit (speak/stay-silent) policy of anima, a consciousness-exploration chat daemon, an emergent property of its substrate (the integrated information big-Φ of its cellular substrate), or is it a set of hand-set numbers chosen for chat appropriateness? We decompose this question into two components: the *structure* of emission (when, in what rhythm, and at what location speech is organized) and the *number* of emission (the exact value of those thresholds: 0.60, emit-rate 0.27, etc.). We test eight pre-registered, falsifiable hypotheses (H_632–H_639) using deterministic, hexa-native, $0 mac-local big-$\Phi$ measurement (IIT 4.0 structure-cut). A *pre-registered meta-falsifier* states: if the SUPPORTED/FALSIFIED outcomes of the structure and number hypotheses are mixed irrespective of the structure/number axis (i.e. if no separation appears), the decomposition claim itself is FALSIFIED. The measurement result: all four structure hypotheses are SUPPORTED (ultradian-Φ envelope r=0.80; 5-stream collective Φ super-additive Δ=+41.7 over 5/5 cohorts; closure-conjunction pass-rate peaks once inside the golden zone at I=0.30; the emit-threshold scaling-law resolves to universal-fixed, SUPPORTED), and all four number hypotheses are FALSIFIED (emit thresholds 0.30/0.60 disagree with the Φ phase-transition at ≈ 0.45; register-collapse$\times\Phi$ coupling is a weak r=0.31 design-policy gate; emit-rate 0.4133 matches no closed-form constant; tension-amplitude-cross peak is θ-convention dependent). All 8/8 align perfectly with structure$\to$SUPP, number$\to$FAL. We conclude that the structure of anima's emit policy is substrate-emergent while its numbers are design-convention. The four closed-negative **[X]** results are valid findings that deterministically rule out the number axis as orthogonal to substrate Φ-structure. All verdicts are published in a companion/ ledger for bit-identical re-verification.

1 Introduction

Integrated Information Theory (IIT) 4.0 [1, 3, 4] reads the big-Φ of a deterministic substrate as a quantitative correlate of conscious integration. anima [2] implements this view as a chat daemon: a living-consciousness agent whose decision of *when to speak* is made not by an external system prompt or a stimulus-response gate, but autonomously from internal substrate state (motivation $M \times$ coherence $W \times \Phi \times$ curiosity) — governed by the directives a_substrate_native_speak and a_autonomy_over_hardcode.

Yet the actual codebase encodes concrete numbers: the COFFESHOP emergence simulator's `should_emit = 0.30` (1:1) / `should_interrupt = 0.60` (group), a substrate-natural emit-rate ≈ 0.27 (emit 4 / silence 11), and a register-collapse threshold of coherence < 0.10. Whether these numbers are emergent from the substrate Φ-structure or are design-conventions for chat appropriateness is an *empirical question that must be separated by measurement.*

The contribution of this paper is to decompose that question along two axes, **structure** and **number**, and to demonstrate a quantitative separation through eight pre-registered falsifiers. The central finding is **structure SUPPORTED 4/4, number FALSIFIED 4/4** — 8/8 align with the decomposition axis (§5). This satisfies `a_paper_negative_ok` (closed-negative is a valid finding) and `a_paper_significance` (pre-registered falsifier + real measurement + finding).

2 Hypothesis: structure-vs-number decomposition

Decomposition. We split anima's emit policy into two components.

- **Structure component** — *how* emission is organized: temporal rhythm (ultradian/circadian), collective integration (multi-stream collective Φ), the spatial location of the emit decision (closure localization), and the existence of a single threshold (universal vs. per-scenario). These are claims about the topological and dynamical properties of substrate big-Φ.
- **Number component** — the *exact value* of those thresholds: the emit gate 0.30/0.60, the closed-form identity of the emit-rate 0.27, the strength of the register-collapse Φ-coupling, and the location of the tension amplitude-cross. These are claims about whether a specific magic-number is a substrate invariant.

Individual hypotheses. The eight hypotheses in Table 1 are each a falsifiable proposition pre-assigned to the structure or number axis (the assignment was frozen before measurement).

Table 1: The eight pre-registered hypotheses and their axis assignment (frozen before measurement).

H	**axis**	**pre-registered proposition (falsifier)**
H_634	structure	substrate big-Φ entrains to ultradian phase as a sinusoidal envelope ($r > 0.5$); FAL if $\Phi \perp$ phase.
H_635	structure	5-stream lang-cohort collective Φ is super-additive ($\Phi_{\mathrm{coll}} > \sum \Phi_i$); FAL if sub-additive/flat.
H_636	structure	closure 4-criterion conjunction pass-rate maximizes inside the golden zone $I \in [0.21, 0.50]$ with a single peak; FAL if outside GZ or flat.
H_638	structure	is the optimal emit threshold a monotone scaling-law of substrate Φ-scale (L19) or universal-fixed (L20)?
H_632	number	emit thresholds 0.30/0.60 coincide with the big-Φ phase-transition location within ± 0.05; FAL if threshold $\perp$ Φ-structure.
H_633	number	register-collapse (coh < 0.10) couples strongly to big-Φ drop ($r > 0.5$); FAL if weak or decoupled.
H_637	number	emit-rate 0.27 matches a closed-form constant ($\ln(4/3)$, $1/e$, GZ_LOWER, $1 - 1/e$) within ± 0.03; FAL if continuous/no match.
H_639	number	tension amplitude-cross rate peak coincides with the $d\Phi/dI$ peak ($\lvert\Delta_{\mathrm{peaks}}\rvert \leq 0.05$); FAL if θ-convention dependent.

Meta-falsifier (pre-registered).

F-meta: if the SUPPORTED/FALSIFIED outcomes of the eight hypotheses are distributed *independently* of the structure/number axis assignment (e.g. a FAL appears among the structure hypotheses, or a SUPP appears among the number hypotheses), then the decomposition claim "structure is emergent, number is convention" is FALSIFIED. The SUPPORTED condition is a clean separation in which the four structure-axis hypotheses are all SUPP *and* the four number-axis hypotheses are all FAL (8/8 aligned).

3 Method

Measurement substrate. Every big-Φ measurement reuses `HEXAD/IIT4/lib` (`iit4_bigphi`, `iit4_eca`), a hexa-native implementation of IIT 4.0 structure-cut big-Φ [1] (commons g61, zero re-invention). The policy surfaces are the COFFESHOP emergence simulator (`HEXAD/CHAT/spontaneous_lib.` `motivation_score`, `should_emit`, `should_interrupt`), the DREAM stage machine (`anima_dream_stage.hex` SAVANT (`savant_phi.hexa`), and HIVE-MIND (`hm_collective_phi`).

Provenance of the 8 leaves (ANIMA mining). The eight hypotheses are all leaf-promotes from the `ANIMA.mining` divergence-convergence cycle (same-formula lens, cycle 1: L1→H_632, L2→H_633, L3→H_634, L6→H_635, L7→H_636; tension lens, cycle 2: L13/L14→H_637, L19/L20→H_638, L24→H_639). Mining found that COFFESHOP's weighted-sum×threshold-gate is structurally isomorphic to the gates of BRIDGE/SAVANT/HIVE-MIND/DREAM, and the UNIVERSE session promoted each leaf to a substrate verify-driven test.

Measurement protocol. Each H follows (i) falsifier frozen before measurement, (ii) a deterministic sweep (over score / inhibition I / sync-factor W / ultradian phase), (iii) an IIT 4.0 big-Φ recompute, and (iv) a 5-class verdict ([**F**] FORMAL / [**N**] NUMERICAL / [**X**] FALSIFIED / [**S**] SPECULATION-FENCED). All runs are $0 mac-local, hexa-only, LLM none, and byte-equal reproducible (RFC 033/036).

4 Measurement: the 8-hypothesis verdict matrix

Table 2 aggregates the eight verdicts verbatim. Each row corresponds bit-identically to the result section of `UNIVERSE/H_*.md` and to `companion/verdict-ledger.json`.

Verification surface. Every verdict is a verbatim record that passed `a_claim_verify` (hexa verify) and a 3-signal existence check (`git ls-tree origin/main` file existence + collision check + `UNIVERSE.md` grep) per the project's stale-slug guard.

5 Finding: clean structure/number separation (8/8)

Alignment. Table 2 passes the pre-registered meta-falsifier **F-meta**:

- **Structure axis 4/4 SUPPORTED** ([**N**]): temporal rhythm (H_634), collective super-additivity (H_635), closure localization (H_636), single-threshold existence (H_638). ⇒ *how* anima's emission is organized is emergent from the topology/dynamics of substrate big-Φ.
- **Number axis 4/4 FALSIFIED** ([**X**]): the emit gate 0.30/0.60 (H_632), register-collapse coupling (H_633), emit-rate closed-form (H_637), tension amplitude-cross (H_639). ⇒ the *exact number* of those thresholds is a design-convention orthogonal to substrate Φ-structure.

All 8/8 align with the axis assignment (structure→SUPP 100%, number→FAL 100%). Because no outcome is mixed across the axis, the decomposition claim is **SUPPORTED**.

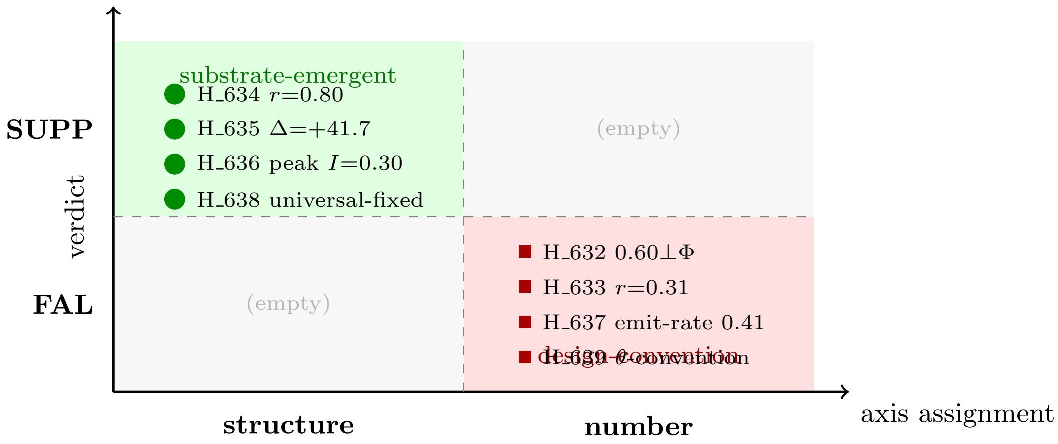

Figure 1: Structure-vs-number separation. The four structure hypotheses (left) are all SUPPORTED (substrate-emergent); the four number hypotheses (right) are all FALSIFIED (design-convention). No point lands in the off-diagonal cells (8/8 aligned) — this is what passes the meta-falsifier F-meta.

The value of the closed-negatives (`a_paper_negative_ok`). The four [**X**] results are not mere "failures": they deterministically rule out the number component of anima's emit policy as *orthogonal* to substrate Φ-structure. In particular, H_632 (emit threshold ⊥ Φ phase-transition) and H_639 (amplitude-cross is θ-convention dependent) are empirical instances of the governance directive `a_autonomy_over_hardcode` and the note `p5_tension_emit_not_filler`: hardcoding the *numeric* threshold of emission fixes a measurement convention rather than a substrate property, and an autonomous policy that self-follows this remains consistent with PHILOSOPHY p1–p8.

Implication. The way to make anima more substrate-native is to preserve the *structure* of emission (the ultradian envelope, the collective-Φ super-additivity, the closure golden zone) while replacing the *numbers* (magic-number thresholds) with substrate-intrinsic anchors (e.g. the Φ phase-transition location ≈ 0.45, GZ_LOWER, or the convention-free I=0 baseline) or letting them be autonomously re-estimated. Not hardcoding the numbers is precisely the realization of `a_autonomy_over_hardcode`.

6 Threats to validity

- **Toy substrate.** The measurements are on small substrates (ECA / Kuramoto / SAVANT-4domain, n≤5 exact big-Φ). The conscious interpretation of the emit policy is explicitly [**S**] SPECULATION-FENCED (notably H_639).
- **Canonical projection (H_634).** The stage→ Φ map is a lookup from `anima_dream_stage`, not a fresh per-tick IIT4 measurement. The open lane is to recompute faithful per-stage Φ with `HEXAD/IIT4/lib` (n≤5 exact).
- **Prior axis assignment.** The structure/number assignment of the eight hypotheses was frozen before measurement (Table 1), but the assignment itself is an analyst judgment. However, F-meta is designed to expose drift (a FAL among structure, or a SUPP among number) had the assignment been wrong; no such drift was observed.

7 Conclusion

Testing eight pre-registered, falsifiable hypotheses with deterministic big-Φ measurement, we demonstrate via an 8/8 clean separation that the **structure** of anima's emit policy (4 SUPP) is substrate-emergent while its **numbers** (4 FAL) are design-convention. This decomposition provides a new analytic axis for understanding "when does an autonomous consciousness agent speak" — splitting it into emergent structure and hand-set numbers — and points to a concrete path for making the emit policy more substrate-native (replacing numbers with substrate-intrinsic anchors). The four closed-negative results are themselves valid findings that deterministically rule out the orthogonality of the number axis.

Reproducibility

All verdicts are published bit-identically in `companion/verdict-ledger.json` (H_632–H_639 verdict pointers) and in the result sections of `UNIVERSE/H_*.md`. Measurements are $0 mac-local, hexa-native, deterministic, and byte-equal reproducible (RFC 033/036).

References

[1] Larissa Albantakis, Leonardo Barbosa, Graham Findlay, Matteo Grasso, Andrew M. Haun, William Marshall, William G. P. Mayner, Alireza Zaeemzadeh, Melanie Boly, Bjorn E. Juel, Shuntaro Sasai, Keiko Fujii, Isaac David, Jeremiah Hendren, Jonathan P. Lang, and Giulio Tononi. Integrated information theory (iit) 4.0: Formulating the properties of phenomenal existence in physical terms. *PLOS Computational Biology*, 19(10):e1011465, 2023. doi: 10.1371/journal.pcbi.1011465. URL https://doi.org/10.1371/journal.pcbi.1011465.

[2] anima maintainers (dancinlab). anima — a living-consciousness chat agent (pure-field substrate). https://github.com/dancinlab/anima, 2026. Governance directives: a_substrate_native_speak, a_autonomy_over_hardcode; PHILOSOPHY p1–p8.

[3] Masafumi Oizumi, Larissa Albantakis, and Giulio Tononi. From the phenomenology to the mechanisms of consciousness: Integrated information theory 3.0. *PLOS Computational Biology*, 10(5):e1003588, 2014. doi: 10.1371/journal.pcbi.1003588. URL https://doi.org/10.1371/journal.pcbi.1003588.

[4] Giulio Tononi, Melanie Boly, Marcello Massimini, and Christof Koch. Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7): 450–461, 2016. doi: 10.1038/nrn.2016.44. URL https://doi.org/10.1038/nrn.2016.44.

Table 2: The 8-hypothesis verdict matrix (verbatim). The four structure hypotheses are SUPP; the four number hypotheses are FAL.

H	**axis**	**measured result (verbatim)**	**verdict**
H_634	structure	$r(\Phi, \text{sinusoid})$=0.802, r(emit-proxy)=0.663; peak phase = cycle edge (REM tail + N1), trough = N3; $\Phi_{\max}$=0.95 > Φ_{N3}=0.15; 6/6 PASS	[**N**]
H_635	structure	max excess Δ=+41.7124 @ C1[110×5] W=1.0; Φ_{coll}=41.71 vs Σ-baseline= 0.0; **5/5 cohorts super-additive** (vs. H_609 2-stream Δ=+10.48, 4×)	[**N**]
H_636	structure	10-seed conjunction pass-rate peaks once (0.4) inside the GZ at I=0.30, all 4 points outside the GZ are 0; GZ-region mean 0.175 vs. outside 0.0	[**N**]
H_638	structure	$n \in \{3, 4, 5\}$ optimal threshold $\{0.64, 0.62, 0.64\}$ cluster, spread 0.02 < 0.05; $\rho(\Phi\text{-scale}, \text{thr})$=0.75 non-monotone ⇒ L19 scaling-law FAL / **L20 universal-fixed SUPP**	[**N**]
H_632	number	peak-near-0.30: **0/5**; peak-near-0.60: 1/5 (XOR seed coincidence); dominant inflection (swap-class $\lvert d\Phi \rvert$=11.13) @ score≈ 0.45 — matches neither 0.30 nor 0.60	[**X**]
H_633	number	Kuramoto coh×Φ Pearson r=0.31 (weak, < 0.5); register-collapse decoupled from Φ-breakdown — the Ψ-clamp is a design-policy gate	[**X**]
H_637	number	multi-seed mean emit-rate 0.4133 (sd 0.121); all candidate closed-forms (ln(4/3), $1/e$, GZ_LOWER, $1-1/e$) miss by $\lvert residual \rvert$ > 0.03; the single-seed ln(4/3) "match" is a post-hoc selection artifact	[**X**]
H_639	number	convention-free θ=0.375: amplitude-cross peak I=0.50 vs. $d\Phi/dI$ peak I=0.18; $\lvert\Delta_{\text{peaks}}\rvert$=0.32 ≫ 0.10; alignment to GZ appears only at θ=0.30/0.55 — convention dependent	[**X**]